SEED-IV Preprocessing & Feature Engineering

Rigorous Data Cleaning, Feature Extraction, & Leakage-Free Validation

SECTION 1 - Dataset Overview & Sample Balance

The SEED-IV dataset consists of preprocessed multi-channel EEG recordings collected from 15 subjects across three sessions. The dataset is used to classify four distinct emotional categories: neutral, sad, fear, and happy. We verify the window-level and subject-level sample balance configurations. Subject contributions are perfectly balanced, with exactly 2,505 windows extracted per subject. The class distributions are reasonably balanced with modest variations (Neutral: 10,170, Sad: 10,245, Fear: 9,225, Happy: 7,935), with Happy samples being somewhat underrepresented due to stimulation duration constraints.

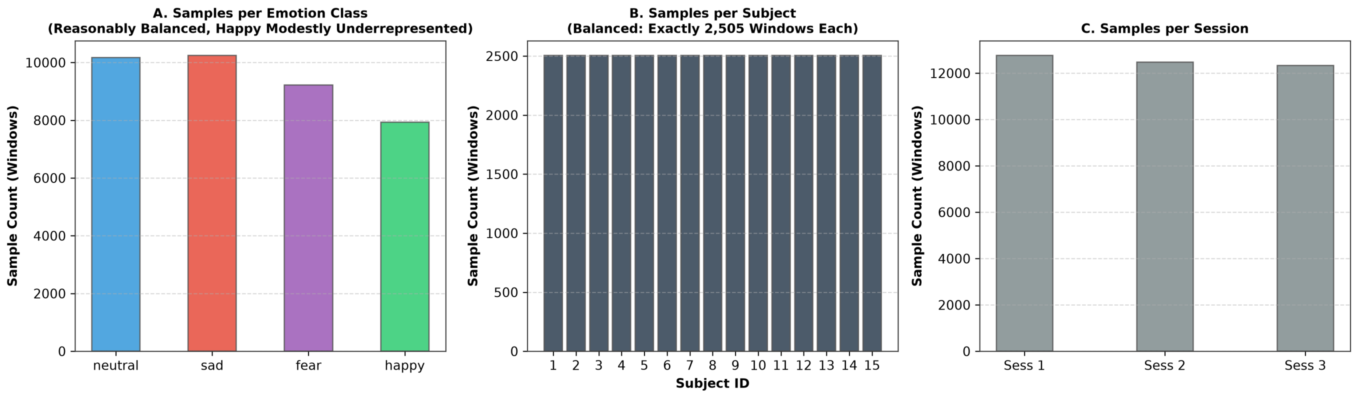

Figure 1: SEED-IV sample distribution configurations across classes, subjects, and sessions.

SECTION 2 - Feature Value Distribution & Standardization

The raw Differential Entropy (DE) feature values exhibit substantial scale and distribution differences across the five frequency bands (Delta, Theta, Alpha, Beta, Gamma) when averaged across the 62 channels. Higher bands show higher power ranges and wider variances. This requires standardizing features to a mean of 0 and standard deviation of 1. To prevent data leakage, scaling parameters are fit strictly on training trials, then transformed on validation/testing data, keeping test characteristics unseen during training.

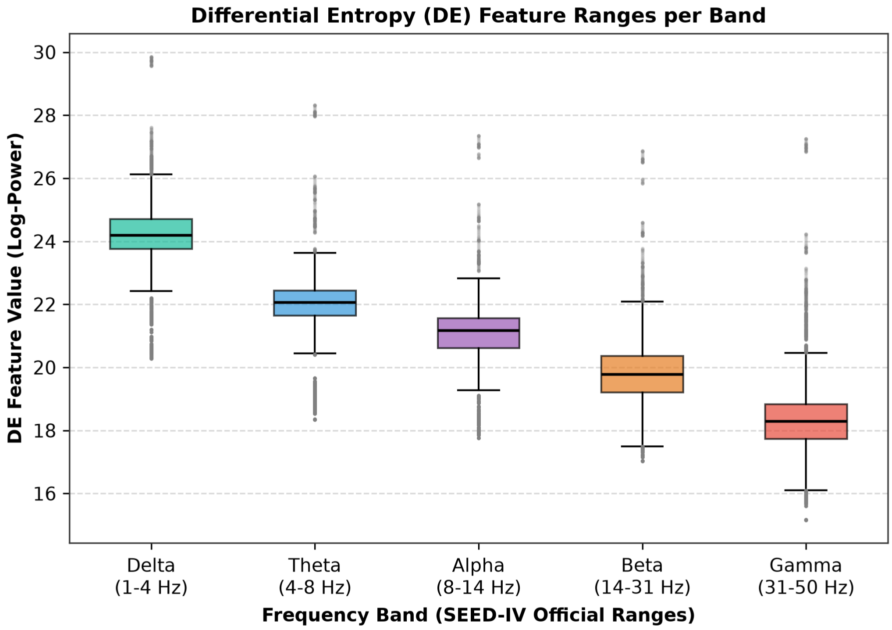

Figure 2: Distribution of raw DE values across the 5 spectral bands.

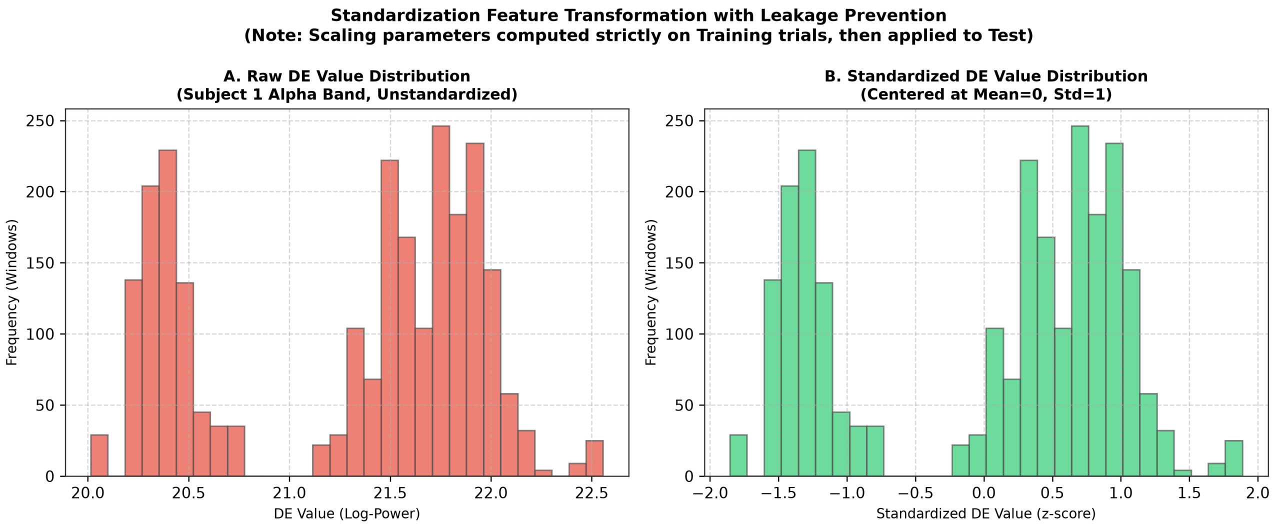

Figure 3: DE feature distribution before (Raw) and after (Standardized) scaling.

SECTION 3 - Feature Extraction Pipeline & Selection

The feature extraction pipeline transforms raw EEG signals into spatial-spectral features. Filtered bands are partitioned into 4-second windows. Differential Entropy is computed for a Gaussian distribution as: DE = 0.5 * log(2 * pi * e * sigma^2). A Linear Dynamical System (LDS) is then applied to smooth the DE values across time. We select the DE + LDS variant over PSD and moving average alternatives, as DE provides logarithmic normalization matching human sensory characteristics, while LDS models temporal dynamics without lag.

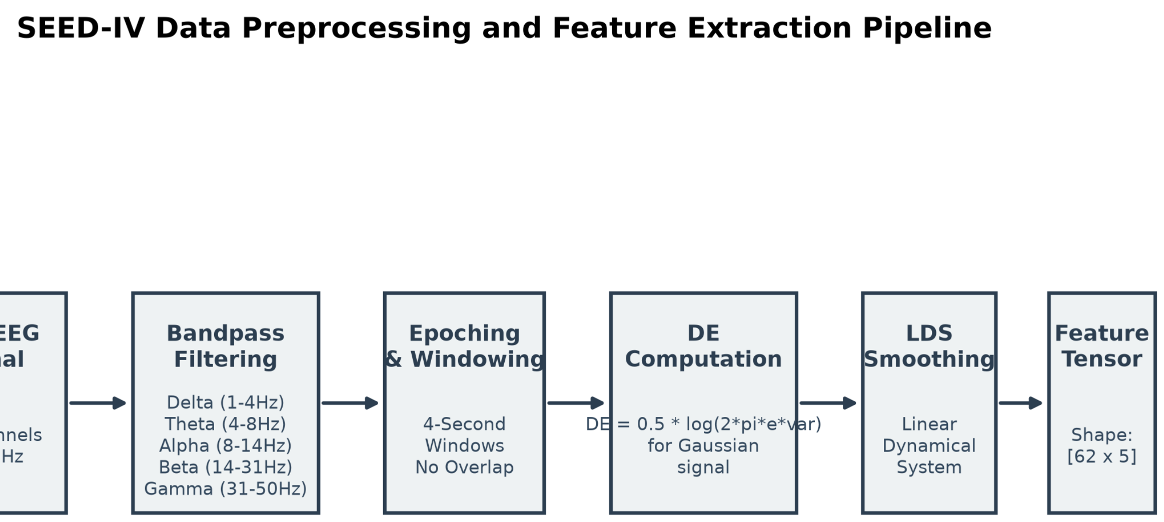

Figure 4: SEED-IV data preprocessing and feature extraction pipeline flowchart.

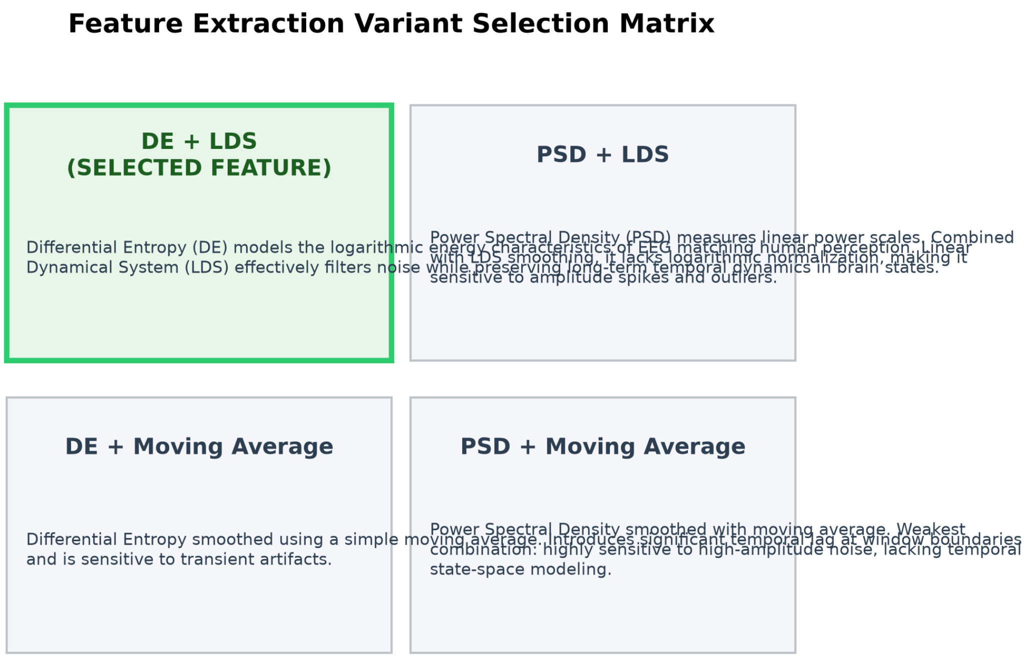

Figure 5: 2x2 grid representing the feature variant selection matrix.

SECTION 4 - Electrode Montage & Input Representation

The spatial arrangement of the 62 electrodes is mapped via a 2D scalp layout. Individual windows represented as 62 channels x 5 bands DE heatmaps capture active spatial-spectral patterns. Correlated channels represent localized brain activations, indicating that a Graph Attention Network (GAT) utilizing physical k-NN adjacencies will effectively model spatial relationships.

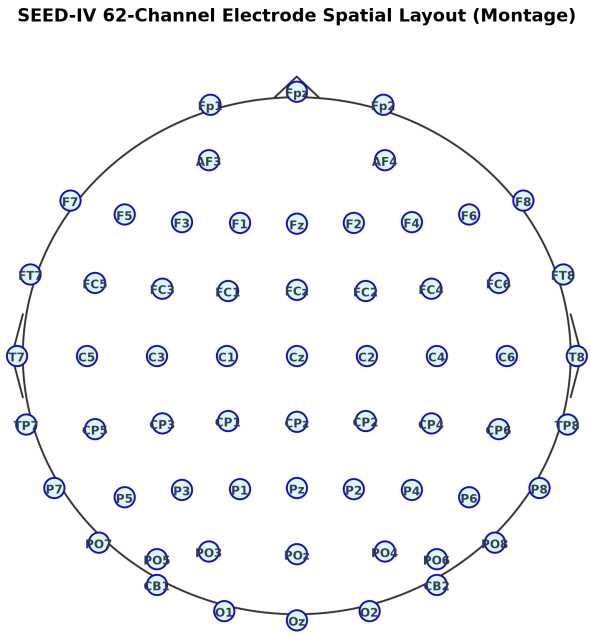

Figure 6: 2D scalp layout projection of the 62 electrode positions.

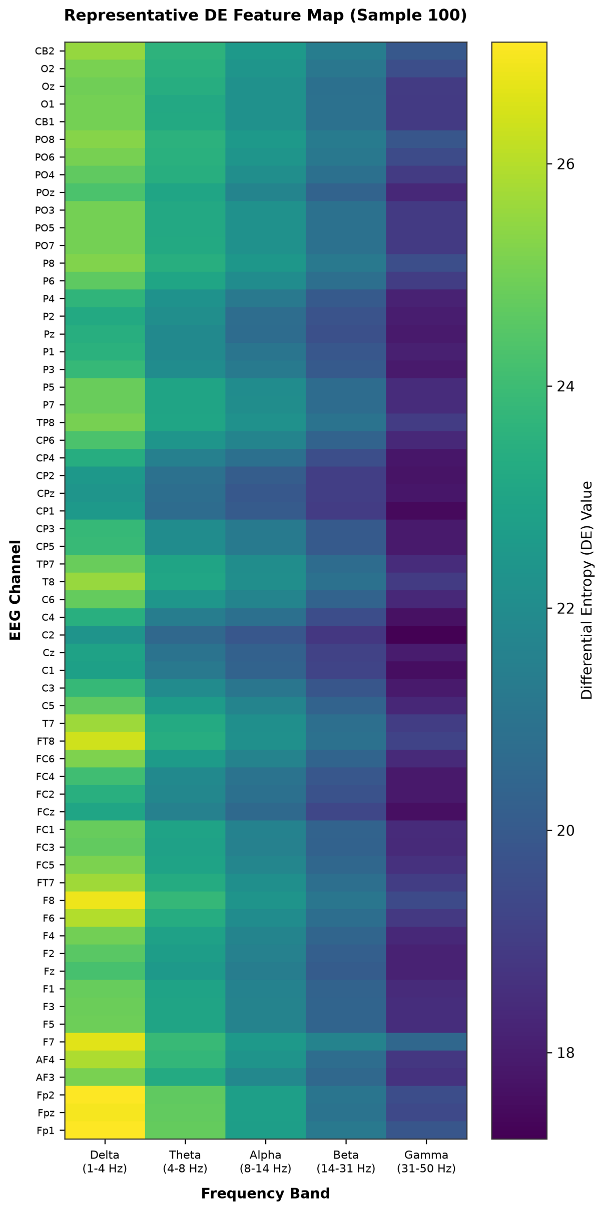

Figure 7: Representative single-window 62x5 DE feature heatmap.

SECTION 5 - Trial-Level Splitting & Durations

Preventing data leakage is essential for obtaining reproducible model evaluations. In sliding-window EEG protocols, adjacent windows share up to 90% overlapping samples. If splitting is performed at the individual window level, near-duplicate samples from the same trial leak across the train and test splits, causing artificially inflated accuracies. To ensure a leakage-free protocol, we enforce trial-level splitting: every window from a given trial is assigned entirely to either the train or test set. This trial duration variance justifies window-level modeling and sliding window sequences over padding or truncating whole trials, which would introduce excessive zero-padding.

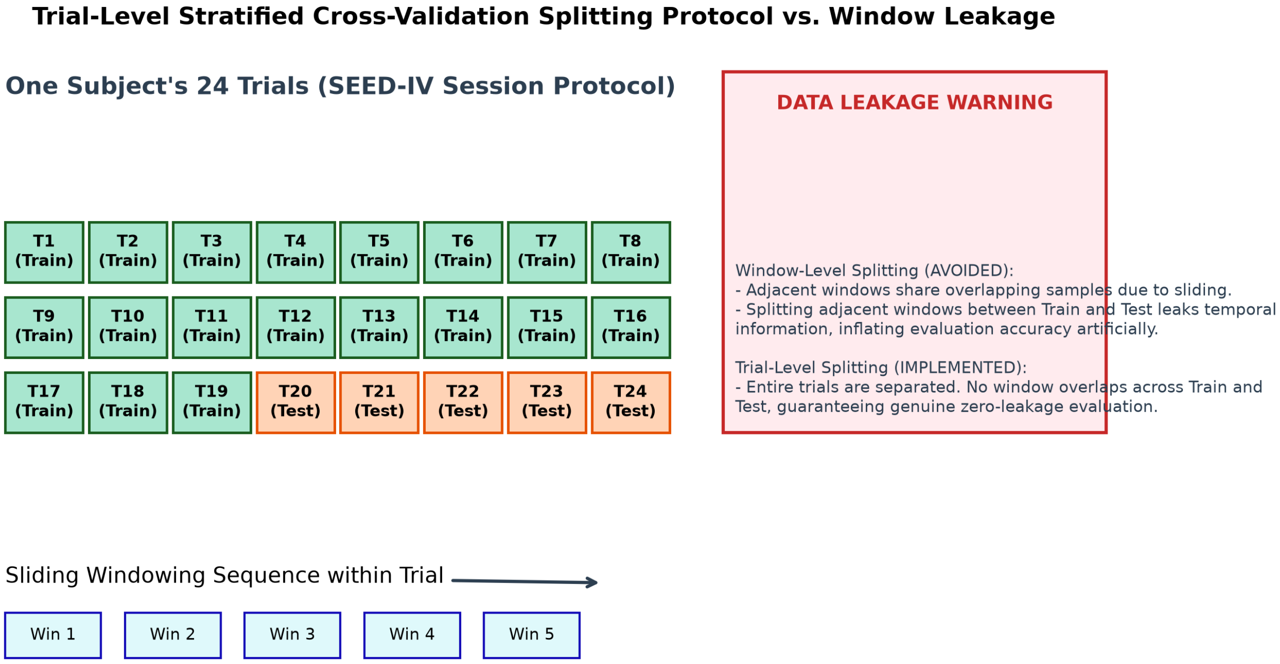

Figure 8: Trial-level stratified splitting protocol vs. window-level leakage.

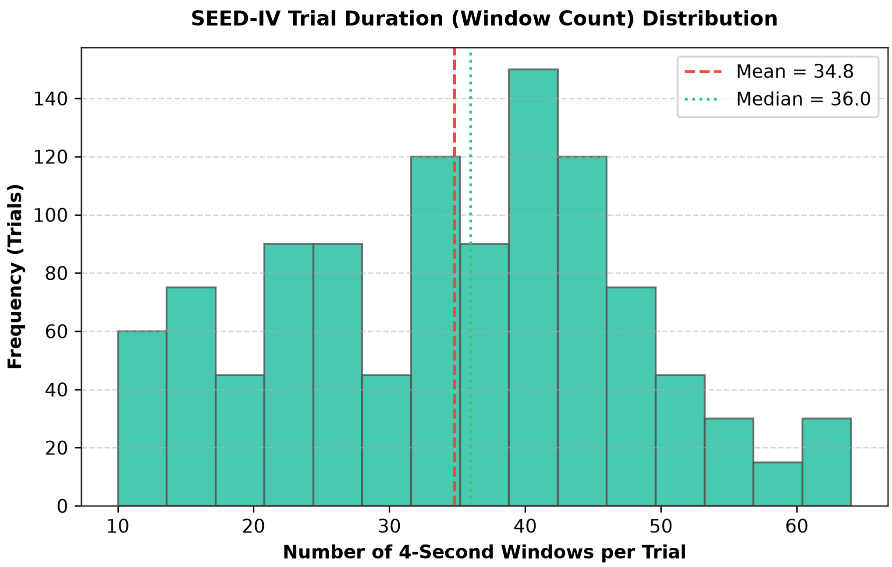

Figure 9: Trial window-count duration distribution histogram.